Woojung Son

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Education

University of Florida — Gainesville, FL
M.S. in Agricultural and Biological Engineering — May 2025 – May 2027 (Expected)
GPA: 3.89/4.0

Chonnam National University — Gwangju, South Korea
B.S. in Biosystems Engineering — Feb. 2021 – Feb. 2025
GPA: 4.32/4.50 | *Summa Cum Laude*

Research Interests

Spatial Reasoning in Multimodal LLMs, Multi-view and 3D Scene Representation, Vision-Language-Action Models

Experience

University of Florida — Gainesville, FL
Graduate Research Assistant — May 2025 – May 2027
USDA Robotics Strawberry Harvesting Project | Advisor: Prof. Won Suk Lee | In collaboration with Purdue University

- Collected the first real-world 6D pose ground truth dataset of strawberries in agricultural fields and constructed a scene-level realistic synthetic dataset using NVIDIA Isaac Sim.
- Evaluated baseline 6D pose estimation models and analyzed the sim-to-real gap in agricultural field conditions.

Chonnam National University — Gwangju, South Korea
Teaching Assistant, Computer Programming — Spring 2024
Assisted undergraduate students in C++, MATLAB, and Simulink.

Publications

[1] **Woojung Son**, Won Suk Lee, Zijing Huang, Daeun Choi, Catia Silva, Yu She, Yan Gu, "From Simulation to the Real-World: An In-Field 6D Pose Dataset and Baseline for Robotic Strawberry Harvesting", *arXiv preprint*, 2026.

Presentations

[1] **Woojung Son** et al., "Geometry-Grounded Transformer-Based Monocular 6D Pose Estimation for Robotic Strawberry Harvesting via Sim-to-Real Transfer", *American Society of Agricultural and Biological Engineers (ASABE) Annual International Meeting*, Indianapolis, IN, 2026. (Oral)

Honors and Awards

Graduate Assistantship (Full Funding), University of Florida — 2025 – 2027

1st Place, AKABFE Student Paper Competition, ASABE Annual International Meeting — 2026

Skills

Programming: Python, C++

Libraries: PyTorch, OpenCV, Open3D, Hugging Face Transformers

Tools: Git, SLURM, LaTeX, NVIDIA Isaac Sim, COLMAP